

untangle::actuator

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Chapter 1

actuator

An actuator is a generic callable that may invoke a list of actions(callables of type `std::function<...>`).

Its goal is to provide a light and simple mechanism to connect components that want to communicate with each other. It works similar as the "signal and slots" mechanisms from other frameworks, but much simplified.

Modern programming languages do not offer class intrinsic interfaces that would improve significantly how objects are connected and how they communicate. Instead, languages, like C++, are using external interfaces, that are not only cluttering the code, but they don't feel "natural" as they are opposed to what an interface is in reality. Basically the objects and interfaces are not separated, and the interface is intrinsic to the object(think on the cogs in a mechanism). A generic callable is an attempt to mitigate this problem. `async` implementation is such use case.

An actuator provides the results of all actions in a vector, in the same order as the actions were added.

Example: how to use actuator instead of polymorphism

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
    for (const auto& s : shapes)
    {
        s->rotate(angle);
    }
}

std::shared_ptr<triangle> t(new triangle);
std::shared_ptr<circle> c(new circle);
std::shared_ptr<square> s(new square);

// using polymorphism
std::vector<shape*> shapes;
shapes.push_back(t.get());
shapes.push_back(c.get());
shapes.push_back(s.get());

rotate_shapes(shapes, 10);

// using actuator
auto action1 = untangle::bind(t, &triangle::rotate);
auto action2 = untangle::bind(c, &circle::rotate);
auto action3 = untangle::bind(s, &square::rotate);
auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

For convenience there are provided helpers methods to "connect" to an initial list of "actions", or to create bindings to class methods.

Please check the manual in [doc/refman.pdf](#) for further references.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [11](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::actuator functor	15
--------------------------	---	--------------------

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::actuator< action_t >	
An actuator is a functor that can trigger a dynamic list of actions (of type std::↵ function<...>)	17
untangle::invalid_action	
Invalid action exception	24

Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

actuator.hpp	27
--	----

Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Functions

- `template<typename action_t, typename... Actions>`
`actuator< action_t > untangle::connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (class_t *obj, T class_t::*method)`
Binding to a class method.

6.1.1 Detailed Description

6.1.2 Function Documentation

6.1.2.1 connect()

```
template<typename action_t, typename... Actions>
actuator< action_t > untangle::connect (
    action_t & A1,
    Actions &... An)
```

Creates an actuator holding an initial list of actions.

Parameters

A1	- The first action. It is specified as <code>std::function<...></code> .
----	--

An	- Any number of further actions, of the same type as A1.
----	--

Returns

An [actuator](#).

Example:

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle) {
    for (const auto& s : shapes) {
        s->rotate(angle);
    }
}

auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

6.1.2.2 bind() [1/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::<
type>>>
action_t untangle::bind (
    const std::shared_ptr< class_t > & obj,
    T class_t::* method)
```

Binding to a class function member.

It returns a `std::function(lambda)` that wraps the function member. It may be used to provide an action for [connect\(\)](#) or [actuator::add\(\)](#).

Remarks

It requires a shared pointer to the class type. A `std::weak_ptr` to it is captured internally in a lambda, so the binding can always check whether the shared object is still alive without keeping it alive itself. Therefore, it is safe to use actions provided by this binding inside an [actuator](#), and safe for the action to outlive the caller's shared pointer.

Parameters

obj	- Class object.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

`action_t` - A `std::function` that wraps the pointer to function member.

Remarks

If the class object gets invalid, invoking this binding will throw an exception of type [invalid_action](#).

6.1.2.3 bind() [2/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↵
type>>
action_t untangle::bind (
    class_t * obj,
    T class_t::* method)
```

Binding to a class method.

Attention

It is not safe to use this binding when the pointed-to object may be destroyed. A null pointer is detected and reported as a dead action, but a pointer to an already destroyed object can not be distinguished from a valid one. Prefer the overload taking a `std::shared_ptr`.

Remarks

It is provided for convenience of use: within a class it is safe to create bindings through this pointer. Binding a null pointer produces an action that is always dead: invoking it throws [invalid_action](#), and an [actuator](#) drops it.

Parameters

obj	- Pointer to class.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

action_t - A `std::function` that wraps the pointer to function member.

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to `untangle::actuator` functor.

Classes

- struct `invalid_action`
Invalid action exception.
- struct `actuator`
An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Functions

- `template<typename... Args>`
`auto last_arg (Args &&... args)`
Get last argument of params pack;.
- `template<typename action_t, typename... Actions>`
`actuator< action_t > connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (class_t *obj, T class_t::*method)`
Binding to a class method.

7.1.1 Detailed Description

Interface to `untangle::actuator` functor.

7.1.2 Function Documentation

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7.1.2.1 last_arg()

Template Parameters

Args	params template types
------	-----------------------

Parameters

args	params
------	--------

Returns

last param value in case it exist, otherwise 0.

Chapter 8

Class Documentation

8.1 untangle::actuator< action_t > Struct Template Reference

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

```
#include <actuator.hpp>
```

Public Types

- using `actions_t` = `std::list<action_t*>`
Actions container type.
- using `results_t` = `std::vector<typename result_t::type>`
Results container type.

Public Member Functions

- `actuator & operator=` (`const actuator &other`)=default
Assignment operator.
- `action_t type` () const
Helper method to access the action type of this object.
- void `reset` ()
Remove all actions and any stored results.
- `template<typename last_t>`
void `invoke_callback` (`last_t &last`)
Invoke the trailing callback argument, if the invocation has one.
- `template<typename... Args>`
void `operator()` (`Args &&... args`)
The call operator.
- `template<typename... Args>`
void `invoke_action` (`const std::string &name, Args &&... args`)
Invokes one single action associated with a key.
- void `add` (`action_t *action`)
Add an action to the actions list.
- void `add` (`const std::string &name, action_t *action`)
Add action to the actions map associated with a name.

- void `remove` (const action_t *action)
Remove an action from the actions list.
- void `remove` (const std::string &name)
Remove an action from actions map.
- bool `is_connected` () const
Check if this actuator is "connected" with other actions.
- bool `has_action` (const std::string &name) const
Check if there is certain named action.

Public Attributes

- `actions_t` actions
Actions list.
- `actions_map_t` actions_map
Named actions map.
- `results_t` results
Actions return values list.

8.1.1 Detailed Description

```
template<typename action_t>
struct untangle::actuator< action_t >
```

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Remarks

An actuator object can be constructed with an initial list of actions by `connect()`.

Callback convention: when the last argument of an invocation can be called with the action return type `R` and returns nothing, it is not passed on as a plain argument alone – it is also treated as a completion callback. The action is invoked with the full argument list as usual, and the callback is then invoked with the action return value. Any void returning callable qualifies: a `std::function<void(R)>`, a lambda taking `R`, a function pointer. It applies to `operator()()` and to `invoke_action()` alike. Actions returning void have no result to report, so no callback is invoked for them.

Warning

The return type is what separates a callback from an argument the action means to consume itself. A trailing callable that returns a value – a transform, a comparator – is left alone, and by the same rule a callback written to return something is silently not invoked. A callback returns nothing.

Template Parameters

<code>action_t</code>	Action type. It is specified as <code>std::function<...></code> .
-----------------------	---

8.1.2 Member Typedef Documentation

8.1.2.1 actions_t

```
template<typename action_t>
using untangle::actuator< action_t >::actions_t = std::list<action_t*>
```

Actions container type.

Remarks

The elements stored are of pointer type, that is required to implement the [remove\(\)](#) operation. `std::function` supports only equality operator for nullptr (two `std::function(s)` can not compare).

8.1.2.2 results_t

```
template<typename action_t>
using untangle::actuator< action_t >::results_t = std::vector<typename result_t::type>
```

Results container type.

It holds the return values of the actions that have a non-void return type. Upon the actuator invocation, the returns can be extracted from [results](#).

8.1.3 Member Function Documentation

8.1.3.1 operator=()

```
template<typename action_t>
actuator & untangle::actuator< action_t >::operator= (
    const actuator< action_t > & other) [default]
```

Assignment operator.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
untangle::actuator<decltype(actuator_rotate.type())> actuator_rotate_1 = actuator_rotate;
actuator_rotate_1(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.2 type()

```
template<typename action_t>
action_t untangle::actuator< action_t >::type () const [inline]
```

Helper method to access the action type of this object.

Note

Intended to be used in expressions by decltype(<[actuator](#) instance>.type())

Returns

action_t

8.1.3.3 reset()

```
template<typename action_t>
void untangle::actuator< action_t >::reset () [inline]
```

Remove all actions and any stored results.

After this call the actuator is empty: [actuator::is_connected\(\)](#) returns false.

8.1.3.4 invoke_callback()

```
template<typename action_t>
template<typename last_t>
void untangle::actuator< action_t >::invoke_callback (
    last_t & last) [inline]
```

Invoke the trailing callback argument, if the invocation has one.

Does nothing unless last_t can be called with the action return type and returns nothing, so any other trailing argument is left alone. The test is on what the type can do, not on what it is, so a plain lambda qualifies as well as a std::function. See the callback convention on [actuator](#).

Remarks

The single condition does the work of two tests: (1) whether last can be called with the action return type at all, and (2) whether that call returns nothing. Spelled out as traits they need two nested if constexpr:

```
// (1)
if constexpr (std::is_invocable_v<last_t&, typename result_t::type>) {
    // (2)
    if constexpr (std::is_void_v<std::invoke_result_t<last_t&, typename result_t::type>») {
        last(results.back\(\));
    }
}
```

Both are folded into the one expression instead, each in a different part of it:

```
requires { requires std::is_void_v<decltype( last(results.back\(\)) )>; }
// \underline{(2)} \underline{(1)}
```

Remarks

(1) Invocability comes for free. To form `decltype(last(results.back()))` the compiler must first form the call `last(...)`. If `last` is an int, that expression does not exist – and inside a `requires-expression` a substitution failure yields `false` instead of being ill formed, the same property concepts are built on. So invocability is never tested separately: it is a precondition of the return type question, and failing it is what makes the whole condition false.

(2) The return type test is the inner `requires`, and the doubled keyword is not a typo. `requires { expr; }` asks only whether `expr` is well formed, and `std::is_void_v<...>` is well formed when it is `false` too – that spelling would let a value returning callable through. The second keyword makes it a nested requirement: a condition that has to hold, not merely compile.

The trait spelling has to nest, which is part of why it is not the one used here: the two `if constexpr` cannot be joined with `&&`. `&&` short circuits evaluation, not instantiation, so `std::invoke_result_t` is formed whatever the first trait says, and for a type that is not invocable at all it has no `::type` – a hard error rather than a false result.

The caller must copy the callback out of the argument pack before invoking the action: an action taking it by value moves from the caller's `std::function`, which would leave this an empty function to call.

Template Parameters

<code>last_</code> <code>_t</code>	Type of the last argument, as returned by <code>last_arg()</code> .
---------------------------------------	---

Parameters

<code>last</code>	- Last argument value, copied before the action was invoked.
-------------------	--

8.1.3.5 operator>()()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::operator() (
    Args &&... args) [inline]
```

The call operator.

Actions in the `actuator::actions` list are triggered by invoking the call operator.

Parameters

<code>args</code>	- Arguments list must match the action arity. A trailing callback is invoked once per action, with that action's return value; see the convention on <code>actuator</code> .
-------------------	--

8.1.3.6 `invoke_action()`

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::invoke_action (
    const std::string & name,
    Args &&... args) [inline]
```

Invokes one single action associated with a key.

Parameters

name	- Key associated with the action.
args	- Arguments list must match the action arity. A trailing callback is invoked with the action's return value; see the convention on actuator .

8.1.3.7 `add()` [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
    action_t * action) [inline]
```

Add an action to the actions list.

Parameters

action	- Action to be added.
--------	-----------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(2).WillRepeatedly(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.add(&action1);
actuator_rotate(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.8 `add()` [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
    const std::string & name,
    action_t * action) [inline]
```

Add action to the actions map associated with a name.

Parameters

name	- Name of the action.
action	- Action to be added.

8.1.3.9 remove() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const action_t * action) [inline]
```

Remove an action from the actions list.

An invalid action (empty std::function) is implicitly removed when `operator()()` is invoked.

Parameters

action	- Action to be removed.
--------	-------------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(0);
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.remove(&action1);
actuator_rotate(50);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.10 remove() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const std::string & name) [inline]
```

Remove an action from actions map.

Parameters

name	- Name of the action to remove.
------	---------------------------------

8.1.3.11 is_connected()

```
template<typename action_t>
bool untangle::actuator< action_t >::is_connected () const [inline]
```

Check if this actuator is "connected" with other actions.

Returns

- true - if the `actuator::actions` list is not empty.
- false - if the `actuator::actions` list is empty.

8.1.3.12 has_action()

```
template<typename action_t>
bool untangle::actuator< action_t >::has_action (
    const std::string & name) const [inline]
```

Check if there is certain named action.

Parameters

name	- Name associated with the action.
------	------------------------------------

Returns

- true - if name can be found in `actuator::actions_map`
- false - if name can not be found in `actuator::actions_map`

The documentation for this struct was generated from the following file:

- `actuator.hpp`

8.2 untangle::invalid_action Struct Reference

Invalid action exception.

```
#include <actuator.hpp>
```

Public Member Functions

- `invalid_action` (std::string text)
Construct a new invalid action object.
- `const char * what () const` noexcept override
The message text describing the reason of this exception.

8.2.1 Detailed Description

Invalid action exception.

Remarks

An action may be provided as a binding to a class function member, by using [untangle::bind\(\)](#). When the class object gets invalid, invoking the action will raise an exception to this type.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 invalid_action()

```
untangle::invalid_action::invalid_action (  
    std::string text) [inline], [explicit]
```

Construct a new invalid action object.

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.2.3 Member Function Documentation

8.2.3.1 what()

```
const char * untangle::invalid_action::what () const [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- actuator.hpp

Chapter 9

File Documentation

9.1 actuator.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <exception>
00009 #include <functional>
00010 #include <iostream>
00011 #include <list>
00012 #include <map>
00013 #include <memory>
00014 #include <string>
00015 #include <type_traits>
00016 #include <utility>
00017 #include <vector>
00018
00019 namespace untangle {
00020 // exception
00029 struct invalid_action : std::exception {
00035     explicit invalid_action(std::string text) : message(std::move(text)) {}
00036
00042     const char* what() const noexcept override { return message.c_str(); }
00043
00044 private:
00045     std::string message;
00046 };
00047
00055 template <typename... Args>
00056 auto last_arg(Args&&... args) {
00057     if constexpr (sizeof...(Args) <= 0) {
00058         return 0;
00059     } else {
00060         return std::get<sizeof...(Args) - 1>(std::forward_as_tuple(args...));
00061     }
00062 }
00063
00085 template <typename action_t>
00086 struct actuator final {
00094     using actions_t = std::list<action_t*>;
00095     using actions_map_t = std::map<std::string, action_t*>;
00096     using result_t = std::conditional<std::is_void<typename action_t::result_type>::value, int,
00097                                     typename action_t::result_type>;
00104     using results_t = std::vector<typename result_t::type>;
00105
00106     actions_t actions;
00107     actions_map_t actions_map;
00108     results_t results;
00109
00110     actuator() = default;
00111     actuator(const actuator&) = default;
00112     actuator(actuator&&) noexcept = default;
00113     ~actuator() = default;
00120     actuator& operator=(const actuator& other) = default;
00121     actuator& operator=(actuator&&) noexcept = default;
00122
00130     action_t type() const { return nullptr; }
00131
00137     void reset() {
```

```

00138     actions.clear();
00139     actions_map.clear();
00140     results.clear();
00141 }
00142
00198 template <typename last_t>
00199 void invoke_callback(last_t& last) {
00200     if constexpr (requires { requires std::is_void_v<decltype(last(results.back()))>; }) {
00201         last(results.back());
00202     }
00203 }
00204
00213 template <typename... Args>
00214 void operator()(Args&&... args) {
00215     results.clear();
00216
00217     // Copied up front, before any action can move the callback out of the argument pack.
00218     auto last = last_arg(args...);
00219
00220     // Dead bindings are collected here and dropped after the loop.
00221     // The actuator does not own the actions it points at, so it must never
00222     // write through action_t* into a std::function belonging to the caller.
00223     std::vector<action_t*> dead_actions;
00224
00225     for (const auto& action : actions) {
00226         // A null pointer or an empty std::function can never be invoked. Calling an
00227         // empty one throws std::bad_function_call, which is not an invalid_action and
00228         // would escape this operator, so drop it instead of invoking it.
00229         if (action == nullptr || !*action) {
00230             dead_actions.push_back(action);
00231             continue;
00232         }
00233
00234         try {
00235             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00236                 (*action)(std::forward<Args>(args)...);
00237             } else {
00238                 results.push_back((*action)(std::forward<Args>(args)...));
00239                 invoke_callback(last);
00240             }
00241         } catch (const invalid_action& ia) {
00242             std::cout << ia.what() << std::endl;
00243             dead_actions.push_back(action);
00244         }
00245     }
00246
00247     for (const auto& dead_action : dead_actions) {
00248         actions.remove(dead_action);
00249     }
00250 }
00251
00259 template <typename... Args>
00260 void invoke_action(const std::string& name, Args&&... args) {
00261     results.clear();
00262
00263     // Copied up front, before the action can move the callback out of the argument pack.
00264     auto last = last_arg(args...);
00265     const auto& it = actions_map.find(name);
00266     if (it != actions_map.end()) {
00267         try {
00268             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00269                 (*it->second)(std::forward<Args>(args)...);
00270             } else {
00271                 results.push_back((*it->second)(std::forward<Args>(args)...));
00272                 invoke_callback(last);
00273             }
00274         } catch (const invalid_action& ia) {
00275             std::cout << ia.what() << std::endl;
00276             actions_map.erase(name);
00277         }
00278     }
00279 }
00280
00289 void add(action_t* action) { actions.push_back(action); }
00290
00297 void add(const std::string& name, action_t* action) { actions_map.emplace(name, action); }
00298
00309 void remove(const action_t* action) {
00310     actions.remove_if([&action](const auto& a) { return (action == a); });
00311 }
00312
00318 void remove(const std::string& name) { actions_map.erase(name); }
00319
00326 bool is_connected() const { return !actions.empty() || !actions_map.empty(); }
00327
00335 bool has_action(const std::string& name) const {
00336     return actions_map.find(name) != actions_map.end();

```

```

00337 }
00338 };
00339
00354 template <typename action_t, typename... Actions>
00355 actuator<action_t> connect(action_t& A1, Actions&... An) {
00356     using actuator_t = untangle::actuator<action_t>;
00357     actuator_t actuator;
00358     actuator.actions = {&A1, &An...};
00359
00360     // remove empty actions
00361     actuator.actions.remove_if([](const auto& action) { return (*action == nullptr); });
00362     return actuator;
00363 }
00364
00365 template <typename actuator_t>
00366 void remove_empty_actions(actuator_t& actuator) {
00367     std::vector<typename actuator_t::actions_map_t::const_iterator> removable_iterators;
00368     for (auto it = actuator.actions_map.begin(); it != actuator.actions_map.end(); ++it) {
00369         if (it->second == nullptr) {
00370             removable_iterators.push_back(it);
00371         }
00372     }
00373     for (auto it : removable_iterators) {
00374         actuator.actions_map.erase(it);
00375     }
00376 }
00377
00378 template <typename key_t, typename action_t, typename... Actions>
00379 actuator<action_t> connect(std::pair<key_t, action_t*> A1, Actions... An) {
00380     using actuator_t = untangle::actuator<action_t>;
00381     actuator_t actuator;
00382     actuator.actions_map = {A1, An...};
00383
00384     // remove empty actions
00385     remove_empty_actions(actuator);
00386     return actuator;
00387 }
00388
00389 // generic helpers to remove const qualifier from a function type,
00390 // for instance const member functions
00391 template <typename T>
00392 struct function_remove_const;
00393
00394 template <typename R, typename... Args>
00395 struct function_remove_const<R(Args...)> {
00396     using type = R(Args...);
00397 };
00398
00399 template <typename R, typename... Args>
00400 struct function_remove_const<R(Args...) const> {
00401     using type = R(Args...);
00402 };
00403
00407
00429 template <typename class_t, typename T,
00430     typename action_t = std::function<typename function_remove_const<T>::type>
00431 action_t bind(const std::shared_ptr<class_t>& obj, T class_t::* method) {
00432     return [wp = std::weak_ptr<class_t>(obj), method](auto&&... args) ->
00433         typename action_t::result_type {
00434             // lock() also keeps the object alive for the duration of the call
00435             const auto obj_ = wp.lock();
00436             if (!obj_) {
00437                 // inform the actuator about dead binding
00438                 throw invalid_action("bind: invalid object");
00439             }
00440             return ((*obj_).*method)(std::forward<decltype(args)>(args)...);
00441         };
00442 }
00443
00464 template <typename class_t, typename T,
00465     typename action_t = std::function<typename function_remove_const<T>::type>
00466 action_t bind(class_t* obj, T class_t::* method) {
00467     return [obj, method](auto&&... args) -> typename action_t::result_type {
00468         if (!obj) {
00469             // inform the actuator about dead binding
00470             throw invalid_action("bind: invalid object");
00471         }
00472         return ((obj)->*method)(std::forward<decltype(args)>(args)...);
00473     };
00474 }
00475
00476 } // namespace untangle

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